

Module 4: Lab 3 - Using Auto Scaling

In this lab, you will create a new Amazon Machine Image (AMI) from an existing Amazon Elastic Compute Cloud (Amazon EC2) instance. You will use that AMI as the basis for defining a system that will scale automatically under increasing loads.

Duration

This lab will require approximately **45 minutes** to complete.

Scenario

In this lab, you will create the scalable web server system shown in the following diagram:



scaling architecture diagram

Objectives

After completing this lab, you will be able to:

- Create a new AMI by using the AWS Command Line Interface (AWS CLI).
- Use Amazon EC2 Auto Scaling to scale up the number of servers available for a specific task when other servers are experiencing heavy load.

The following components are created for you as a part of the lab environment:

- Virtual private cloud (VPC)
- Public subnets
- Private subnets
- Amazon EC2 - Command Host (*in the public subnet*): You will log in to this instance to create a few of your AWS assets.

You will create the following components for this lab:

- Amazon EC2 web server
- AMI
- Auto Scaling launch template
- Auto Scaling group
- Auto Scaling policies
- Application Load Balancer

Accessing the AWS Management Console

1. At the top of these instructions, choose Start Lab to launch your lab.

A Start Lab panel opens displaying the lab status.

2. Wait until you see the message "**Lab status: ready**", then choose the **X** to close the Start Lab panel.

3. At the top of these instructions, choose AWS

This will open the AWS Management Console in a new browser tab. The system will automatically log you in.

Tip: If a new browser tab does not open, there will typically be a banner or icon at the top of your browser indicating that your browser is preventing the site from opening pop-up windows. Choose the banner or icon and choose "Allow pop ups."

4. Arrange the AWS Management Console tab so that it displays along side these instructions. Ideally, you will be able to see both browser tabs at the same time, to make it easier to follow the lab steps.

⚠ Please do not change the Region during this lab.

Task 1: Create a New AMI for Amazon EC2 Auto Scaling

In this task, you will launch a new EC2 instance and then create a new AMI based on that running instance. You will use the AWS CLI tools on the Command Host to perform all of these operations.

The following instructions vary slightly depending on whether you are using Windows or Mac/Linux.

Windows Users: Using SSH to Connect

💬 These instructions are for Windows users only.

If you are using macOS or Linux, skip to the next section.

5. In the AWS Management Console, on the **Services**  menu, choose **EC2**.
6. In the left navigation pane, choose **Instances**.

7. Select the **Command Host** instance.
8. From the **Details** tab in the lower pane, copy the **Public IPv4 address** value.
9. Read through the bullet points in this step before completing the actions, because you will not be able to see these instructions when the **Details** panel is open.
 - Choose  above the instructions you are currently reading, and then choose . The Credentials window opens.
 - Choose **Download PPK**, and save the **labsuser.ppk** file. Typically, your browser will save it to the Downloads directory.
 - To exit the **Details** panel, choose the **X** icon.
10. Download needed software.
 - You will use **PuTTY** to SSH to the EC2 instances. If you do not have PuTTY installed on your computer, download it here (<https://the.earth.li/~sgtatham/putty/latest/w64/putty.exe>).
11. Open **putty.exe**.
12. Configure PuTTY to not time out:
 - Choose **Connection**.
 - Set **Seconds between keepalives** to 30

This allows you to keep the PuTTY session open for a longer period.

13. Configure your PuTTY session:
 - Choose **Session**
 - **Host Name (or IP address)**: Paste the *Public DNS or IPv4 address* of the Bastion Host instance that you noted earlier.
 - Back in PuTTY, in the **Connection** list, expand  **SSH**
 - Choose **Auth** and expand  **Credentials**
 - Under **Private key file for authentication**: Choose **Browse**
 - Browse to the *labsuser.ppk* file that you downloaded, select it, and choose **Open**
 - Choose **Open** again
14. To trust and connect to the host, choose **Accept**.
15. When prompted to **login as**, enter `ec2-user`

This will connect you to the EC2 instance.

16. Windows Users: Skip ahead to the next task.

Mac and Linux Users

These instructions are for Mac/Linux users only. If you are a Windows user, skip ahead to the next task.

17. In the AWS Management Console, on the **Services** ▾ menu, choose **EC2**.
18. In the left navigation pane, choose **Instances**.
19. Select the **Command Host** instance.
20. From the **Details** tab in the lower pane, copy the **Public IPv4 address** value.
21. Read through the bullet points in this step before completing the actions, because you will not be able to see these instructions when the **Details** panel is open.
 - Choose  above the instructions you are currently reading, and then choose . The Credentials window opens.
 - Choose **Download PEM**, and save the **labsuser.pem** file.
 - To exit the **Details** panel, choose the **X** icon.
22. Open a terminal window, and change directory `cd` to the directory where the `labsuser.pem` file was downloaded.

For example, run the following command if the file was saved to your Downloads directory:

```
cd ~/Downloads
```

23. Change the permissions on the key to be read only:

```
chmod 400 labsuser.pem
```

24. Return to the terminal window and run the following command. Replace **<public-ip>** with the **Public IPv4** value you copied to your clipboard earlier in the lab:

```
ssh -i labsuser.pem ec2-user@<public-ip>
```

25. When prompted to allow a first connection to this remote SSH server, enter `yes`

Because you are using a key pair for authentication, you will not be prompted for a password.

Create a New EC2 Instance

Now that you are logged in to Command Host, you will use the AWS CLI to create a new instance that hosts a web server.

26. Inspect the script `UserData.txt`, which was installed for you as part of the Command Host.

```
more UserData.txt
```

This script performs a number of initialization tasks, including updating all installed software on the box and installing a small PHP web application that you can use to simulate a high CPU load on the instance. Near the bottom of the script, you see the following lines:

```
find -wholename /root/.*history -wholename /home/*/.*histo
find / -name 'authorized_keys' -exec rm -f {} \;
rm -rf /var/lib/cloud/data/scripts/*
```

These lines erase any history or security information that might have accidentally been left on the instance when the image was taken.

27. Choose **Details** above the instructions you are currently reading, and then choose **Show**.

From the table that displays, copy all the rows into a text file.

28. In the following command, replace **<Amild>**, **<HTTPAccess>**, and **<SubnetId>** with the values you retrieved in the previous step. Then, run the command:

```
aws ec2 run-instances --key-name vockey --instance-type t2
```

The output of this command provides you with an **InstanceId**. This value is referred to as **NEW-INSTANCE-ID** in subsequent steps and should be replaced appropriately.

29. Use the **aws ec2 wait instance-running** command to monitor this instance's status. Replace **<NEW-INSTANCE-ID>** with the value you copied previously.

```
aws ec2 wait instance-running --instance-ids <NEW-INSTANCE
```

Before proceeding to the next step, wait for the command to return to a prompt.

30. Your instance should have started a new web server. To test that the web server was installed properly, run the following command to obtain the public DNS name. Replace **<NEW-INSTANCE-ID>** with the value you copied previously.

```
aws ec2 describe-instances --instance-id <NEW-INSTANCE-ID>
```

Copy the output of this value (without the quotation marks). This value is referred to as **public-dns-address** in the next procedure.

31. Use a web browser to navigate to the following page. Replace **PUBLIC-DNS-ADDRESS** with the value you copied in the last step.

⚠ Note:

- It could take a few minutes for the web server to be installed. Wait for 5 minutes before trying other steps.
- Do not choose **Start Stress** at this stage.

```
http://PUBLIC-DNS-ADDRESS/index.php
```

If your web server does not appear to be running, check with your instructor.

Create a Custom AMI

In this procedure, you will create a new AMI from the EC2 instance you just created.

32. Use the **aws ec2 create-image** AWS CLI command to create a new AMI, based on the new instance you just created. Replace **<NEW-INSTANCE-ID>** with the actual Instance ID of the WebServerBaseImage instance you just created.

```
aws ec2 create-image --name WebServer --instance-id <NEW-I
```

💡 By default, **create-image** will restart the current instance before creating the AMI, to ensure the integrity of the image on the file system. While your AMI is being created, proceed to the next section.

Task 2: Create an Auto Scaling Environment

In this section, you will create a load balancer that pools a group of EC2 instances under a single DNS address. You will use Auto Scaling to create a dynamically scalable pool of EC2 instances based on the image that you created in the previous section.

Finally, you will create a set of alarms that will scale out or scale in the number of instances in your load balancer group whenever the CPU performance of any machine within the group exceeds or falls below a set of specified thresholds.

You could perform the following task by using either the AWS CLI or the AWS Management Console. For simplicity, you will use the console.

Create an Application Load Balancer

In this section, you will create an Application Load Balancer, as shown in the following architecture diagram:



Load Balancing

33. On the **Services** ▼ menu, choose **EC2**.
34. In the left navigation pane, choose **Load Balancers**. (You might need to scroll down to find it.)
35. Choose **Create Load Balancer**
36. Under **Application Load Balancer**, choose **Create**.
37. For **Load balancer name**, enter `webserverloadbalancer`
38. Scroll down to the **Network mapping** section, and configure:
 - For **VPC**, select **Lab VPC**.

You will now specify which *subnets* the load balancer should use. The load balancer will be internet facing, so you will select both public subnets.
 - Choose the *first* displayed Availability Zone. From the **Subnet** menu that displays, choose **Public Subnet 1**.
 - Choose the *second* displayed Availability Zone. From the **Subnet** menu that displays, choose **Public Subnet 2**.

You should now have two subnets selected: **Public Subnet 1** and **Public Subnet 2**. (If not, try the configuration again.)
39. In the **Security groups** section, configure:
 - Choose **Security groups**, and select ☒ **HTTPAccess**.
 - Below the dropdown menu, choose the **X** icon next to the default security group to remove it.

The HTTPAccess security group should now be the only one that appears.

Next, you will create a target group.

Analysis: A *target group* defines where to *send* traffic that comes into the load balancer. The Application Load Balancer can send traffic to multiple target groups based on the URL of the incoming request, such as having requests from mobile apps go to a different set of servers. Your web application will use only one target group.

40. In the **Listeners and routing** section, configure:

- Choose **Create target group**. A new browser tab opens.
- **Choose a target type:** Choose **Instances**.
- **Target group name**, enter `webserver-app`
- Scroll down and expand ► **Advanced health check settings**.

Note: The Application Load Balancer automatically performs *health checks* on all instances to ensure that they are responding to requests. The default settings are recommended, but you will make them slightly faster for use in this lab.

- **Health check path:** Enter `/index.php`
- **Healthy threshold:** Enter `2`
- **Interval:** Enter `10` (seconds)

This means that the health check will be performed every 10 seconds. If the instance responds correctly twice in a row, it will be considered healthy.

- Choose **Next**. The *Register targets* screen appears.

Note: *Targets* are the individual instances that will respond to requests from the load balancer.

You do not have any web application instances yet, so you can skip this step.

- Review the settings, and choose **Create target group**.

41. Return to the browser tab where you already started defining the Load balancer.

42. In the **Listeners and routing** section, choose the refresh icon.

43. For the **Listener HTTP:80** row, set the Default action to forward to **webserver-app**.

44. Scroll to the bottom and choose **Create load balancer**.

The load balancer is successfully created.

45. Choose **View load balancer**.

Create a Launch Template

Your Auto Scaling group will use the launch template to know which AMI to use to create new EC2 instances. For this example, you will launch the AMI that you created previously, which automatically configures itself as a web server when it is launched.

46. In the left navigation pane, choose **Launch Templates**.
47. Choose **Create launch template**
48. For **Launch template name**, enter `WebServerLaunchTemplate`
49. In the **AMI machine image (AMI)** pane, search for `WebServer`
50. Select **WebServer**. It is under the **My AMIs** group.
51. In the **Instance type** section, select **t2.micro**.
52. In the **Network settings** section, for **Security groups**, select **HTTPAccess**.
53. Expand the **Advanced details** section, and set **Detailed CloudWatch monitoring** to **Enable**.
54. Scroll to the bottom, and choose **Create launch template**
- A success message appears.
55. Choose **View launch templates**

Create an Auto Scaling Group

Your Auto Scaling group will create a minimum number of EC2 instances that will reside behind your load balancer. In subsequent procedures, you will also add scale-out and scale-in policies that increase or decrease the number of running instances in reaction to alarms from Amazon CloudWatch.

56. In the left navigation pane, scroll to the bottom of the menu, and choose **Auto Scaling Groups**.
57. Choose **Create Auto Scaling group**
58. In Step 1, **Choose launch template or configuration**, configure:
 - **Auto Scaling group name**: Enter `WebServersASGroup`
 - **Launch template**: Choose **WebServerLaunchTemplate**
59. Choose **Next**
60. In the **Network** pane, configure:
 - **VPC**: Choose **Lab VPC**
 - **Subnets**: Choose **Private Subnet 1** and **Private Subnet 2**
61. Choose **Next**
62. In the **Load balancing** pane, choose **Attach to an existing load balancer**.

63. In the **Attach to an existing load balancer** pane, for **Existing load balancer target groups**, choose **webserver-app**.
64. In the **Additional settings** pane, select ☒ **Enable group metrics collection within CloudWatch**.
65. Choose **Next**

Analysis: In the next few steps, you will configure Auto Scaling to target 45 percent CPU Utilization. If average CPU Utilization exceeds this target, additional instances will be launched. If average CPU Utilization falls below this target, instances *might* be terminated.
66. In the **Group size** pane, configure:
 - **Desired capacity:** Enter `2`
 - **Minimum capacity:** Enter `2`
 - **Maximum capacity:** Enter `4`
67. In the **Scaling policies** pane, choose **Target tracking scaling policy**, and configure:
 - **Scaling policy name:** Enter `MyScalingPolicy`
 - **Metric type:** Choose **Average CPU utilization**
 - **Target value:** Enter `45`
68. Choose **Next**
69. On the **Add notifications** page, choose **Next**
70. On the **Add tags** page, choose **Add tag** and configure:
 - **Key:** Enter `Name`
 - **Value:** Enter `WebApp`
71. Choose **Next**
72. At the bottom of the **Review** page, choose **Create Auto Scaling group**

Verifying the Auto Scaling Configuration

In this task, you will verify that both the Auto Scaling configuration and the load balancer are working. You will use a pre-installed script on one of your servers that will consume CPU cycles, which will invoke the scale-out alarm.

73. In the left navigation pane, choose **Instances**.
74. Verify that two new instances named **WebApp** are being created as part of your Auto Scaling group.

Wait for the two new instances to complete initialization before you proceed to the next step.

Tip: Observe the **Status check** column for the instances until it shows that both status checks have completed successfully.

75. In the left navigation pane, choose **Target Groups**.

76. Select your target group (**webserver-app**).

77. In the **Targets** tab in the lower half of your screen, verify that two instances are being created. Refresh this list until the **Status** of these instances changes to **healthy**.

You can now test the web application by accessing it through the load balancer.

78. In the left navigation pane, choose **Load Balancers**.

79. Select ☒ **webserverloadbalancer**.

80. From the **Description** tab below, copy the **DNS name** value.

The value looks like *webserverloadbalancer-xxxxxxxxx.xx-xxxx-x.elb.amazonaws.com*.

This value will be referred to as **load-balancer-url** in the next step.

81. Open a new web browser tab, paste the URL into the address bar, and then press Enter.

82. On the web page, choose **Start Stress**.

This calls the application **stress** in the background, which causes the CPU utilization on the instance that serviced this request to spike to 100 percent.

83. Return to the console, and in the left navigation pane, choose **Auto Scaling Groups**.

84. Select ☒ **WebServerASGroup**.

85. Choose the **Activity** tab for your Auto Scaling group, and scroll down to the **Activity History** pane.

After a few minutes, you should see a new entry, which indicates that your Auto Scaling group is launching a new EC2 instance.

💡 This is because CloudWatch detected that the average CPU utilization of your Auto Scaling group exceeded 45 percent, and your scale-up policy has been invoked in response.

Lab Complete

Congratulations! You have completed the lab.

85. Choose at the top of this page and then choose **Yes** to confirm that you want to end the lab.

A panel will appear, indicating that "DELETE has been initiated... You may close this message box now."

86. Choose the **X** in the top right corner to close the panel.

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